

The Relational Spin-Closure Hypothesis

Mathematical baseline, open problems, and falsification criteria

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Abstract

This document formulates a hypothesis and a falsification programme. Its mathematical baseline is the companion paper *A Conditional Construction of Lorentzian Geometry from Euclidean Three-Space: Clifford Double-Cover Data, Gluing, and a Soldering Isomorphism* [1]. Starting from the positive-definite Euclidean space

$$V = \mathbb{R}^3,$$

that paper constructs

$$\mathcal{S} = \mathbb{R} \oplus V, \quad N(a, v) = a^2 - \langle v, v \rangle, \quad \text{Cl}_3 = \text{Cl}(V, q),$$

a Clifford norm-one group G_{Spin} , a proper orthochronous Lorentz group $G_{\text{Lor}} \subset O(B)$, and a central two-fold covering

$$1 \longrightarrow \{\pm 1\} \longrightarrow G_{\text{Spin}} \xrightarrow{\rho} G_{\text{Lor}} \longrightarrow 1.$$

It also constructs the orthogonal Lie algebra

$$\mathfrak{so}(B) \cong \Lambda^2 \mathcal{S}, \quad \dim \mathfrak{so}(B) = 6.$$

Global Lorentzian geometry requires additional data. A gluing datum yields a smooth four-manifold only under explicit topological and smoothness hypotheses. Independently, a G_{Spin} -valued Čech 1-cocycle $\tilde{g}_{ij}$ projects to

$$g_{ij} = \rho(\tilde{g}_{ij}).$$

Local soldering data A_i satisfy

$$A_j(\phi_{ij}(y)) = D\phi_{ij}(y) A_i(y) g_{ij}(x_y),$$

and determine a vector-bundle isomorphism

$$s : E_g \xrightarrow{\sim} TM.$$

The induced Lorentzian metric is

$$g_x(X, Y) = B(s_x^{-1}X, s_x^{-1}Y).$$

The existence of such soldering data is not a consequence of the Euclidean input or of the group-valued cocycle alone.

For one explicit periodic one-parameter family, every finite parameter admits soldering data; any two parameters are gauge equivalent under G_{Spin} -valued Čech 0-cochains; and corresponding soldering data induce the same Lorentzian metric. No general statement about arbitrary cocycle families follows.

The next open mathematical problem is the infinitesimal bridge

$$\mathrm{Lie}(G_{\mathrm{Spin}}) \stackrel{?}{\cong} \mathfrak{so}(B), \quad d\rho_e : \mathrm{Lie}(G_{\mathrm{Spin}}) \stackrel{?}{\longrightarrow} \mathfrak{so}(B).$$

Connections, parallel transport, holonomy, curvature and torsion lie beyond this open step. The Relational Spin-Closure Hypothesis begins only after these distinctions: it proposes that a future classical configuration space carries a gauge-equivariant equation whose solution set couples state and geometric variables nontrivially. Quantization and particle interpretations are later tests, not premises.

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1 Status convention

The labels used below have the following meanings:

P	proved or constructed in Paper I;
S	standard external mathematics;
O	open mathematical problem;
H	hypothesis introduced here;
T	later mathematical or empirical benchmark.

No statement labelled O, H or T is asserted as a theorem of Paper I.

2 Mathematical results used from Paper I

2.1 Local algebraic construction

The primitive local datum is

$$V = \mathbb{R}^3, \quad \langle u, v \rangle > 0 \quad (u \neq 0). \quad (2.1)$$

No manifold, tangent bundle, Lorentzian metric, orientation, principal bundle or connection is part of this datum.

Paper I constructs the real spin factor

$$\mathcal{S} = \mathbb{R} \oplus V, \quad N(a, v) = a^2 - \langle v, v \rangle, \quad (2.2)$$

with polarized bilinear form B of signature $(1, 3)$, together with the real Clifford algebra

$$\text{Cl}_3 = \text{Cl}(V, q), \quad q(v) = \langle v, v \rangle. \quad (2.3)$$

The Clifford norm-one group G_{Spin} acts on the paravector model and yields

$$1 \longrightarrow \{\pm 1\} \longrightarrow G_{\text{Spin}} \xrightarrow{\rho} G_{\text{Lor}} \longrightarrow 1, \quad (2.4)$$

where $G_{\text{Lor}} \subset O(B)$ is the proper orthochronous Lorentz group defined by the determinant and future-cone conditions used in Paper I. The homomorphism ρ is continuous and admits local continuous sections. No smooth Lie-group structure on G_{Spin} is asserted there.

The B -skew-adjoint endomorphisms form

$$\mathfrak{so}(B) = \{A \in \text{End}_{\mathbb{R}}(\mathcal{S}) : B(Ax, y) + B(x, Ay) = 0\}, \quad (2.5)$$

and Paper I proves a linear isomorphism

$$\Lambda^2 \mathcal{S} \xrightarrow{\sim} \mathfrak{so}(B), \quad \dim_{\mathbb{R}} \mathfrak{so}(B) = 6. \quad (2.6)$$

The identification

$$\text{Lie}(G_{\text{Spin}}) \cong \mathfrak{so}(B) \quad (2.7)$$

is *not* a result of Paper I.

2.2 Global quotient and tangent transitions

Let $\{D_i\}_{i \in I}$ be local four-dimensional pieces. A gluing datum specifies overlap domains and homeomorphisms

$$\phi_{ij} : W_{ij} \longrightarrow W_{ji} \quad (2.8)$$

satisfying the stated identity, inverse and cocycle conditions. Under the closed-gluing-relation and countability hypotheses of Paper I, the quotient is Hausdorff and second countable. If the transition maps are smooth, the resulting atlas defines a smooth four-manifold M .

The local pieces alone do not determine a unique quotient:

$$\{D_i\}_{i \in I} \not\Rightarrow \text{a unique } M. \quad (2.9)$$

The tangent transition cocycle is

$$T_{ij}(y) = D\phi_{ij}(y). \quad (2.10)$$

Independently, let

$$\tilde{g}_{ij} : U_i \cap U_j \longrightarrow G_{\text{Spin}} \quad (2.11)$$

be a G_{Spin} -valued Čech 1-cocycle, and set

$$g_{ij} := \rho \circ \tilde{g}_{ij} : U_i \cap U_j \longrightarrow G_{\text{Lor}}. \quad (2.12)$$

In general,

$$D\phi_{ij} \neq g_{ij}. \quad (2.13)$$

Thus the tangent and Lorentz-valued transition cocycles are distinct data before an additional compatibility condition is imposed.

2.3 Soldering isomorphism and induced Lorentzian metric

Let $E_g \rightarrow M$ be the Lorentz vector bundle defined by the cocycle g_{ij} . Local soldering data are smooth maps

$$A_i : D_i \longrightarrow \text{GL}(\mathcal{S}) \quad (2.14)$$

satisfying

$$A_j(\phi_{ij}(y)) = D\phi_{ij}(y) A_i(y) g_{ij}(x_y). \quad (2.15)$$

They determine a vector-bundle isomorphism

$$s : E_g \xrightarrow{\sim} TM. \quad (2.16)$$

The inverse

$$s^{-1} : TM \xrightarrow{\sim} E_g \quad (2.17)$$

has the conventional direction of an E_g -valued solder form.

The induced metric

$$g_x(X, Y) := B\left(s_x^{-1}X, s_x^{-1}Y\right) \quad (2.18)$$

is a smooth Lorentzian metric on M .

The logical dependence is therefore

$$\boxed{(V, \langle \cdot, \cdot \rangle) \Longrightarrow (\mathcal{S}, B, \text{Cl}_3, G_{\text{Spin}}, \rho, \mathfrak{so}(B))} \quad (2.19)$$

locally, but globally

$$\boxed{\text{gluing datum} + \tilde{g} + \text{soldering data} \Longrightarrow (M, TM, g).} \quad (2.20)$$

Neither additional term on the left of (2.20) is derived solely from (2.19).

2.4 Lifting obstruction on the chosen cover

For a Lorentz-valued Čech 1-cocycle g_{ij} , assume continuous lifts

$$h_{ij} : U_i \cap U_j \longrightarrow G_{\text{Spin}}, \quad \rho(h_{ij}) = g_{ij} \quad (2.21)$$

have been chosen on the double overlaps. On triple overlaps define

$$\epsilon_{ijk} := h_{jk} h_{ij} h_{ik}^{-1} \in \{\pm 1\}. \quad (2.22)$$

This defines the $\mathbb{Z}/2$ -valued Čech 2-cocycle class used in Paper I. The paper does not identify this class with the classical characteristic class $w_2(TM)$. Likewise the determinant condition obtained from (2.15) is not identified there with $w_1(TM)$.

In particular, the following identifications are not part of the mathematical baseline:

$$[\epsilon] = w_2(TM), \quad [\det D\phi] = w_1(TM). \quad (2.23)$$

2.5 Examples and nonuniqueness

Paper I establishes the following limitations:

$$\text{topological gluing} \not\Rightarrow \text{smooth gluing}, \quad (2.24a)$$

$$\text{pointwise fibre isomorphisms} \not\Rightarrow \text{continuous soldering}, \quad (2.24b)$$

$$\text{orientation-reversing two-chart gluing} \Rightarrow \text{no soldering data}, \quad (2.24c)$$

$$\text{combinatorial labels} \not\Rightarrow \text{existence of soldering data}. \quad (2.24d)$$

On the chosen periodic cover, distinct lifts may have the same Lorentz projection; this is a statement about the chosen cover and the specified equivalence relation, not a computation of $H^1(M; \mathbb{Z}/2)$.

2.6 One-parameter Čech 1-cocycle family

For the explicit periodic family of Paper I, let $\tilde{g}^{(\lambda)}$ denote the G_{Spin} -valued Čech 1-cocycle and let

$$\mathcal{E}_\lambda := \{\text{local soldering data for } \tilde{g}^{(\lambda)}\}. \quad (2.25)$$

Then

$$\forall \lambda \in \mathbb{R}, \quad \mathcal{E}_\lambda \neq \emptyset. \quad (2.26)$$

Moreover, for all $\lambda_1, \lambda_2 \in \mathbb{R}$, there are continuous G_{Spin} -valued Čech 0-cochains u_i such that

$$\tilde{g}_{ij}^{(\lambda_2)} = u_i \tilde{g}_{ij}^{(\lambda_1)} u_j^{-1}, \quad (2.27)$$

and there is an explicit bijection

$$\mathcal{T}_{\lambda_1, \lambda_2} : \mathcal{E}_{\lambda_1} \xrightarrow{\sim} \mathcal{E}_{\lambda_2} \quad (2.28)$$

with

$$g_{\mathcal{T}_{\lambda_1, \lambda_2}}(A) = g_A. \quad (2.29)$$

No statement quantified over arbitrary covers, arbitrary cocycle families or arbitrary one-parameter subgroups follows from (2.26)–(2.29).

3 The Relational Spin-Closure Hypothesis

The constructions above determine the present mathematical boundary. The hypothesis begins beyond that boundary.

3.1 Configuration space and gauge symmetry

The future classical variables have not yet been fully constructed. Accordingly, their final product structure is not assumed. Let

$$\tilde{\mathcal{Q}} \tag{3.1}$$

denote a future space of admissible classical configurations extending the geometric structures already constructed, and let $\mathcal{G}$ be its gauge group. If a quotient is mathematically well behaved, write

$$\mathcal{Q} := \tilde{\mathcal{Q}}/\mathcal{G}. \tag{3.2}$$

No manifold structure on $\mathcal{Q}$ is assumed at this stage.

Hypothesis 3.1 (Relational Spin-Closure Hypothesis). *There exist a classical configuration space $\tilde{\mathcal{Q}}$, a gauge group $\mathcal{G}$, a target space $\mathcal{Y}$, and a map*

$$\mathfrak{C} : \tilde{\mathcal{Q}} \longrightarrow \mathcal{Y} \tag{3.3}$$

such that:

(i) $\mathfrak{C}$ is gauge equivariant and $0 \in \mathcal{Y}$ is fixed by the gauge action;

(ii) the physical solution set

$$\mathcal{Z} := \mathfrak{C}^{-1}(0)/\mathcal{G} \tag{3.4}$$

is nonempty and contains nontrivial gauge classes;

(iii) the geometric variables entering $\tilde{\mathcal{Q}}$ are solved jointly with the state variables rather than prescribed as fixed external functions;

(iv) the local Euclidean/Clifford/Lorentz construction of Paper I is not altered in order to obtain a desired macroscopic equation.

The notation $\mathfrak{C}$ names the proposed equation only. It is not identified with torsion, curvature, a Čech cocycle, a connection, or any previously constructed object.

3.2 Reciprocity

Suppose a future differentiable formulation admits a local splitting

$$T_q \tilde{\mathcal{Q}} = X_\Phi \oplus X_G, \tag{3.5}$$

where X_Φ denotes state variations and X_G denotes geometric variations. At a solution $q \in \mathfrak{C}^{-1}(0)$, write

$$D\mathfrak{C}_q(\delta\Phi, \delta G) = 0. \tag{3.6}$$

Definition 3.2 (Linearized reciprocity). *A solution q is linearly reciprocal if*

$$\ker D\mathfrak{C}_q \cap ((X_\Phi \setminus \{0\}) \times (X_G \setminus \{0\})) \neq \emptyset. \tag{3.7}$$

Thus the linearized solution space contains at least one nontrivial variation in which both sectors vary.

A stronger local condition may be imposed on specified subspaces when the partial linearizations

$$C_\Phi := D_\Phi \mathfrak{C}_q, \quad C_G := D_G \mathfrak{C}_q \tag{3.8}$$

are invertible there. Then

$$\delta G = -C_G^{-1} C_\Phi \delta\Phi, \quad \delta\Phi = -C_\Phi^{-1} C_G \delta G. \tag{3.9}$$

Equation (3.9) is a possible regularity condition, not part of the present mathematical baseline.

3.3 Classical-first restriction

The hypothesis is classical at its defining stage. No Hilbert-space state axiom, Born rule, canonical commutation or anticommutation relation, Fock-space construction, path integral, or quantum-statistical postulate is used to define $\mathfrak{C}$.

No clock scalar, lapse, preferred foliation or simultaneity convention belongs to the primitive data. If such an object is introduced later, it requires an independent definition and status.

A variational realization, if it exists, would be a stronger statement:

$$\mathcal{A} : \tilde{\mathcal{Q}} \longrightarrow \mathbb{R}, \quad D\mathcal{A} = 0 \implies \mathfrak{C} = 0. \quad (3.10)$$

No particular action functional is assumed.

4 Open mathematical problems before a connection theory

4.1 The infinitesimal Spin–Lorentz bridge

Paper I provides the group-level map $\rho : G_{\text{Spin}} \rightarrow G_{\text{Lor}}$ and the Lie algebra $\mathfrak{so}(B)$, but not the missing infinitesimal identification. The next open statements are

$$\text{Lie}(G_{\text{Spin}}) \stackrel{?}{\cong} \mathfrak{so}(B), \quad (4.1)$$

and

$$d\rho_e : \text{Lie}(G_{\text{Spin}}) \stackrel{?}{\longrightarrow} \mathfrak{so}(B). \quad (4.2)$$

A smooth Lie-group structure on G_{Spin} must be supplied before (4.1)–(4.2) can be interpreted literally in the standard sense.

An intermediate algebraic construction may be studied first. For Clifford conjugation c , the tangent condition suggested by $gc(g) = 1$ at the identity is

$$X + c(X) = 0. \quad (4.3)$$

A future formalization may define the corresponding linear subspace of Cl_3 , construct its infinitesimal action on $\mathcal{S}$, and compare that action with $\mathfrak{so}(B)$. Until the relevant comparison theorem is proved, this subspace is not denoted $\text{Lie}(G_{\text{Spin}})$.

4.2 Connection, curvature and torsion

Only after an infinitesimal bridge is available does the standard connection-level architecture become applicable. If $P \rightarrow M$ is a principal G_{Spin} -bundle with a principal connection ω , then standard differential geometry gives the curvature

$$F_\omega = d\omega + \frac{1}{2}[\omega \wedge \omega]. \quad (4.4)$$

With a solder form $e : TM \rightarrow E_g$, torsion is

$$T = D_\omega e. \quad (4.5)$$

These are standard definitions [8, 3]; neither object has yet been constructed in the present formal development.

A group homomorphism cannot be applied directly to a Lie-algebra-valued one-form. In particular,

$$\rho \circ \omega \quad (4.6)$$

is not a definition of a Lorentz connection. Projection of a connection requires the infinitesimal map $d\rho_e$ or an explicitly proved equivalent representation.

5 Finite graph model

A discrete model may be used before the continuum connection theory is complete, provided its observables are not identified with continuum objects by name alone.

Let

$$\Gamma = (\mathcal{V}, \mathcal{E}) \quad (5.1)$$

be an oriented graph. Assign

$$U_{xy} \in G_{\text{Spin}}, \quad U_{yx} = U_{xy}^{-1} \quad (5.2)$$

to each oriented edge. A vertex gauge transformation $h_x \in G_{\text{Spin}}$ acts by

$$U_{xy} \mapsto h_y U_{xy} h_x^{-1}. \quad (5.3)$$

For a closed path $C = (x_0, \dots, x_n = x_0)$, define

$$H_C := U_{x_{n-1}x_n} \cdots U_{x_0x_1}. \quad (5.4)$$

Then

$$H_C \mapsto h_{x_0} H_C h_{x_0}^{-1}, \quad (5.5)$$

so its conjugacy class $[H_C]$ is gauge invariant.

The object $[H_C]$ is a graph observable. No equality

$$[H_C] = [\text{Hol}_\omega(C)] \quad (5.6)$$

is asserted before a continuum-limit and comparison theorem is constructed.

6 Falsification criteria and later benchmarks

6.1 Mathematical falsification criteria

The following implications are excluded by the current mathematics:

$$V = \mathbb{R}^3 \not\Rightarrow M, \quad (6.1a)$$

$$(B, \tilde{g}) \not\Rightarrow \exists s : E_g \xrightarrow{\sim} TM, \quad (6.1b)$$

$$\text{group-level } \rho \not\Rightarrow d\rho_e, \quad (6.1c)$$

$$\prod_{e \in C} U_e \not\equiv \text{Hol}_\omega(C) \quad \text{without a comparison theorem.} \quad (6.1d)$$

Criterion 6.1 (Nontriviality modulo gauge). *A proposed closure variable fails if every member of the tested family belongs to one gauge orbit and all observables used by the proposal are constant on that orbit.*

The one-parameter family (2.25)–(2.29) is an explicit example of such a negative result at the transition-cocycle level. It does not imply the same conclusion for connection-level families.

Criterion 6.2 (No target equation as a definition). *A candidate $\mathfrak{C}$ does not establish RSCH if it is defined by inserting the Einstein equation, a prescribed Schwarzschild field, a geodesic equation, Maxwell/Yang–Mills equations, an externally prescribed redshift/lapse profile, or a desired particle mass as a primitive constitutive law.*

Criterion 6.3 (Scale provenance). *Every dimensionful quantity must have an explicit mathematical source: boundary or initial data, a dimensionful parameter, a normalization of a classical solution, or a later quantum/renormalization mechanism after quantization has independently been justified.*

6.2 Benchmark hierarchy

Required mathematical structure	Later comparison	Status
Classical state space and evolution law	Lorentz-covariant flat-sector dynamics	T
Connection and a mathematically defined flat sector	special-relativistic observables	T
Coupled non-flat classical solutions	weak-field gravitational phenomenology	T
Stable localized finite-energy solutions	energy, momentum and invariant mass	T
Reduced phase space / Hamiltonian or symplectic structure	possible quantization	T
Consistent quantized physical modes	particle/statistics/radiation interpretation	T
Independent internal non-Abelian gauge sector	confinement / QCD comparisons	T

No row may be used as an assumption to prove an earlier row.

7 Research order

The present dependency order is

$$\boxed{\text{P}} \quad (V, \langle \cdot, \cdot \rangle) \implies (\mathcal{S}, B, \text{Cl}_3, G_{\text{Spin}}, \rho, \mathfrak{so}(B)), \quad \text{gluing datum} + \tilde{g} + \text{soldering data} \implies (M, TM, g). \quad (7.1)$$

This is the mathematical content supplied by Paper I.

The first open layer is

$$\boxed{\text{O}_1} \quad \text{Lie}(G_{\text{Spin}}) \stackrel{?}{\cong} \mathfrak{so}(B), \quad d\rho_e. \quad (7.2)$$

The second open layer is

$$\boxed{\text{O}_2} \quad \omega, \quad \text{parallel transport}, \quad \text{Hol}_\omega, \quad F_\omega, \quad T. \quad (7.3)$$

Only after the variables needed for a classical configuration space have been constructed does the hypothesis become a concrete mathematical problem:

$$\boxed{\text{H}} \quad (\tilde{\mathcal{Q}}, \mathcal{G}, \mathcal{Y}, \mathfrak{C}), \quad \mathcal{Z} = \mathfrak{C}^{-1}(0)/\mathcal{G}. \quad (7.4)$$

Physical comparisons are subsequent tests:

$$\boxed{\text{T}} \quad \text{SRT}, \quad \text{gravity}, \quad \text{localized states}, \quad \text{quantization}, \quad \text{strong-sector tests}. \quad (7.5)$$

Thus

$$\boxed{\text{P} \longrightarrow \text{O}_1 \longrightarrow \text{O}_2 \longrightarrow \text{H} \longrightarrow \text{T}.} \quad (7.6)$$

8 Object and status table

Object or statement	Mathematical form	Status
Euclidean input	$V = \mathbb{R}^3$ with positive-definite inner product	P
Real spin factor	$\mathcal{S} = \mathbb{R} \oplus V$, $N(a, v) = a^2 - \langle v, v \rangle$	P
Real Clifford algebra	$\text{Cl}_3 = \text{Cl}(V, q)$	P
Clifford norm-one group	G_{Spin} as defined in Paper I	P
Central two-fold covering	$1 \rightarrow \{\pm 1\} \rightarrow G_{\text{Spin}} \xrightarrow{\rho} G_{\text{Lor}} \rightarrow 1$	P
Orthogonal Lie algebra	$\mathfrak{so}(B) \cong \Lambda^2 \mathcal{S}$, dimension 6	P
Smooth four-manifold	obtained from additional gluing data and smoothness hypotheses	P
Tangent transition cocycle	$D\phi_{ij}$	P
Group-valued Čech 1-cocycle	$\tilde{g}_{ij} : U_i \cap U_j \rightarrow G_{\text{Spin}}$	P
Lorentz-valued cocycle	$g_{ij} = \rho(\tilde{g}_{ij})$	P
Soldering isomorphism	$s : E_g \xrightarrow{\sim} TM$, when local soldering data exist	P
Lorentzian metric	$g_x(X, Y) = B(s_x^{-1}X, s_x^{-1}Y)$	P
Čech lifting class on $\mathcal{U}$	$[\epsilon] \in \check{H}^2(\mathcal{U}; \mathbb{Z}/2)$ for chosen lifts	P
Identification with $w_2(TM)$	not proved in Paper I	O
Identification with $w_1(TM)$	not proved in Paper I	O
Lie algebra of G_{Spin}	$\text{Lie}(G_{\text{Spin}})$	O
Infinitesimal covering map	$d\rho_e$	O
Principal connection	ω	O
Curvature and torsion	$F_\omega, T = D_\omega e$	O
Classical configuration space	$(\tilde{\mathcal{Q}}, \mathcal{G})$	H
Closure equation	$\mathfrak{C} : \tilde{\mathcal{Q}} \rightarrow \mathcal{Y}$, $\mathfrak{C} = 0$	H
Reciprocity	mixed state/geometric directions in $\ker D\mathfrak{C}$	H
Quantum theory	conditional later extension	T

9 Relation to established mathematics

The algebraic terminology follows standard Clifford and spinor literature [4, 5]. Smooth manifolds, vector bundles, principal bundles, connections, solder forms, curvature and torsion are used in their standard differential-geometric meanings [6, 7, 8, 3]. No claim of novelty is attached to those structures.

The specific new proposal of this paper is restricted to the existence and physical adequacy of a future classical equation $\mathfrak{C} = 0$ with the properties stated in Section 3. General relativity, gauge theory, soliton models and non-Abelian strong-interaction theory are comparison targets only [9, 10, 11, 12, 13].

10 Conclusion

The current mathematical construction ends at

$$\begin{aligned}
V = \mathbb{R}^3 &\implies (\mathcal{S}, B, \text{Cl}_3, G_{\text{Spin}}, \rho, \mathfrak{so}(B)), \\
&+ \text{gluing datum} \implies M, TM, \\
&+ \tilde{g}_{ij}, A_i \implies s : E_g \xrightarrow{\sim} TM \implies (TM, g).
\end{aligned} \tag{10.1}$$

The next mathematical questions are

$$\text{Lie}(G_{\text{Spin}}) \stackrel{?}{\cong} \mathfrak{so}(B), \quad d\rho_e, \tag{10.2}$$

followed, if successful, by connection-level geometry.

The hypothesis is the later existence of

$$(\tilde{\mathcal{Q}}, \mathcal{G}, \mathcal{Y}, \mathfrak{C}), \quad \mathfrak{C} : \tilde{\mathcal{Q}} \rightarrow \mathcal{Y}, \quad \mathfrak{C}^{-1}(0)/\mathcal{G} \neq \emptyset, \quad (10.3)$$

with nontrivial gauge classes and reciprocal coupling between the state and geometric sectors. Failure to construct such a nontrivial classical equation, or failure of its later benchmarks, falsifies the corresponding RSCH claim. Nothing in Paper I by itself establishes (10.3).

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